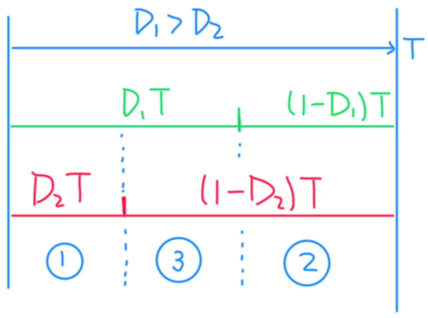
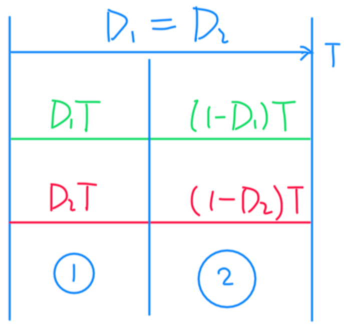


1 D_2T

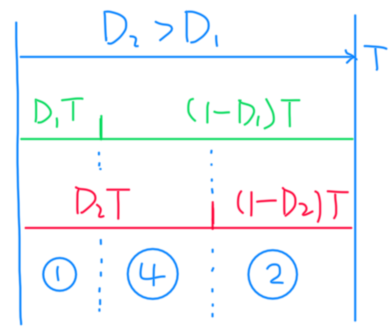
3 $(D_1-D_2)T$

2 $(1-D_1)T$



1 D_1T

2 $(1-D_1)T$



1 D_1T

4 $(D_2-D_1)T$

2 $(1-D_2)T$

$$\Sigma R = R_{B_1}+R_{B_2}+R_{B_3}+R_o$$

1 D_1T & D_2T

$$\begin{bmatrix}\dot{i}_{L_1}\\ \dot{i}_{L_2}\end{bmatrix}=\begin{bmatrix}-\left[\frac{R_{B_2}(\Sigma R-R_{B_2})+\Sigma R(R_S+R_L)}{L_1\Sigma R}\right] & \frac{R_{B_2}R_{B_3}}{L_1\Sigma R}\\ \frac{R_{B_2}R_{B_3}}{L_2\Sigma R} & -\left[\frac{R_{B_3}(\Sigma R-R_{B_3})+\Sigma R(R_S+R_L)}{L_2\Sigma R}\right]\end{bmatrix}\begin{bmatrix}i_{L_1}\\ i_{L_2}\end{bmatrix}+\begin{bmatrix}\frac{-R_{B_2}}{L_1\Sigma R} & \frac{\Sigma R-R_{B_2}}{L_1\Sigma R} & \frac{-R_{B_2}}{L_1\Sigma R}\\ \frac{-R_{B_3}}{L_2\Sigma R} & \frac{-R_{B_3}}{L_2\Sigma R} & \frac{\Sigma R-R_{B_3}}{L_2\Sigma R}\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix}$$

2 $(1-D_1)T$ & $(1-D_2)T$

$$\begin{bmatrix}\dot{i}_{L_1}\\ \dot{i}_{L_2}\end{bmatrix}=\begin{bmatrix}-\left[\frac{R_{B_1}(\Sigma R-R_{B_1})+\Sigma R(R_S+R_L)}{L_1\Sigma R}\right] & \frac{R_{B_1}R_{B_2}}{L_1\Sigma R}\\ \frac{R_{B_1}R_{B_2}}{L_2\Sigma R} & -\left[\frac{R_{B_2}(\Sigma R-R_{B_2})+\Sigma R(R_S+R_L)}{L_2\Sigma R}\right]\end{bmatrix}\begin{bmatrix}i_{L_1}\\ i_{L_2}\end{bmatrix}+\begin{bmatrix}\frac{R_{B_1}-\Sigma R}{L_1\Sigma R} & \frac{R_{B_1}}{L_1\Sigma R} & \frac{R_{B_1}}{L_1\Sigma R}\\ \frac{R_{B_2}}{L_2\Sigma R} & \frac{R_{B_2}-\Sigma R}{L_2\Sigma R} & \frac{R_{B_2}}{L_2\Sigma R}\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix}$$

3 D_1T & $(1-D_2)T$

$$\begin{bmatrix}\dot{i}_{L_1}\\ \dot{i}_{L_2}\end{bmatrix}=\begin{bmatrix}-\left[\frac{R_{B_2}(\Sigma R-R_{B_2})+\Sigma R(R_S+R_L)}{L_1\Sigma R}\right] & -\frac{R_{B_2}(R_{B_2}-\Sigma R)}{L_1\Sigma R}\\ -\frac{R_{B_2}(R_{B_2}-\Sigma R)}{L_2\Sigma R} & -\left[\frac{R_{B_2}(\Sigma R-R_{B_2})+\Sigma R(R_S+R_L)}{L_2\Sigma R}\right]\end{bmatrix}\begin{bmatrix}i_{L_1}\\ i_{L_2}\end{bmatrix}+\begin{bmatrix}\frac{-R_{B_2}}{L_1\Sigma R} & \frac{\Sigma R-R_{B_2}}{L_1\Sigma R} & \frac{-R_{B_2}}{L_1\Sigma R}\\ \frac{R_{B_2}}{L_2\Sigma R} & \frac{R_{B_2}-\Sigma R}{L_2\Sigma R} & \frac{R_{B_2}}{L_2\Sigma R}\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix}$$

4 $(1-D_1)T$ & D_2T

$$\begin{bmatrix}\dot{i}_{L_1}\\ \end{bmatrix}=\begin{bmatrix}-\left[\frac{R_{B_1}(\Sigma R-R_{B_1})+\Sigma R(R_S+R_L)}{L_1\Sigma R}\right] & -\frac{R_{B_1}R_{B_3}}{L_1\Sigma R}\end{bmatrix}\begin{bmatrix}i_{L_1}\\ \end{bmatrix}+\begin{bmatrix}\frac{R_{B_1}-\Sigma R}{L_1\Sigma R} & \frac{R_{B_1}}{L_1\Sigma R} & \frac{R_{B_1}}{L_1\Sigma R}\end{bmatrix}\begin{bmatrix}V_1\\ V_2\end{bmatrix}$$

$$\begin{bmatrix} \dot{i}_{L_2} \end{bmatrix} \quad \begin{bmatrix} -\dfrac{R_{B_1}R_{B_3}}{L_2\Sigma R} & & -\left[\dfrac{R_{B_3}(\Sigma R - R_{B_3}) + \Sigma R(R_S + R_L)}{L_2\Sigma R}\right] \end{bmatrix}\begin{bmatrix} i_{L_2} \end{bmatrix} \quad \begin{bmatrix} \dfrac{-R_{B_3}}{L_2\Sigma R} & \dfrac{-R_{B_3}}{L_2\Sigma R} & \dfrac{\Sigma R - R_{B_3}}{L_2\Sigma R} \end{bmatrix}\begin{bmatrix} v_2 \\ V_3 \end{bmatrix}$$